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14 August 2026 (Friday)

NERVE SYSTEM

Unit → Neuron

→ function → To maintain control and coordination.
→ Endocrine system helps in the process.
→ Study of nerve → Neurology
→ Doctor → Neurologist
→ The nerve system is mainly of two types;

Nerve system

Central Nervous system (C.N.S)
- Brain
- Spinal cord

Peripheral nervous (P.N.S) system
- Nerve emerging from the brain and spinal cord.
- Cranial nerve - 12 pair
- Spinal nerve - 33 pair

→ The peripheral Nervous system is of two types;

Somatic Nervous system
→ The nerves of this system connect to our voluntary muscles and regulates voluntary actions

Autonomous nervous system
→ The nerves of this system connect to our involuntary muscles and internal organs, and they regulate involuntary actions.

→ The autonomous nervous system is of two types :

Sympathetic
→ source of distress
→ Active during emergency situations

Parasympathetic
→ Source of happy-ness.
→ Active after emergency situations end.

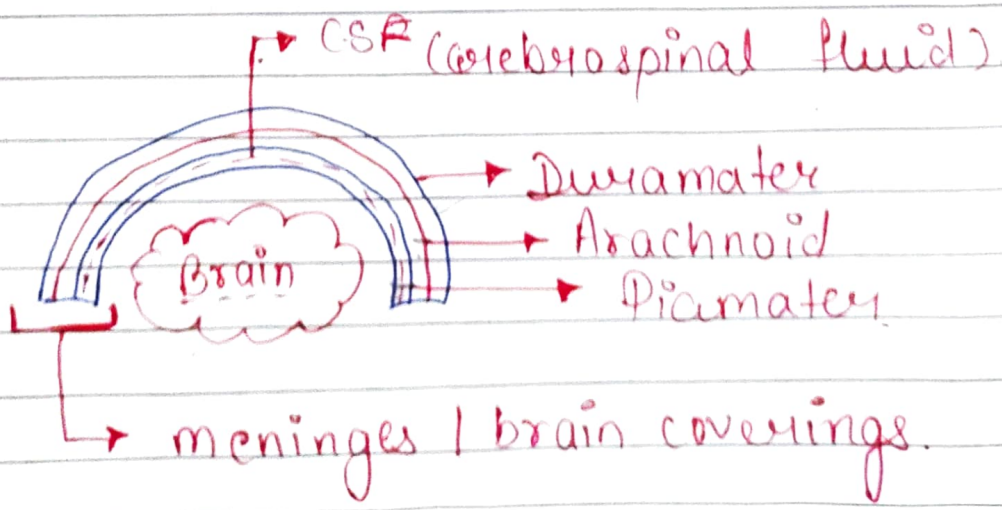
① CENTRAL NERVE SYSTEM

(i) Brain

→ It works to enable our thinking and understanding, remembering, emotions, and to operate and control all organs of the body.

→ Weight → 1300 - 1400 gm.
→ Our brain contains folds
→ A three layered covering protects the brain. These layers are called meninges

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→ If there is a viral or bacterial infection in the brain coverings (menings) → 'Meningitis' (Brain fever).

→ C.S.F (cerebrospinal fluid) is filled between the arachnoid and piamater.

→ Function of C.S.F → (i) Keeping the brain in buoyant state.
(ii) Delivering nutrients to the brain
(iii) Removing waste and excretory products.
(iv) Protecting the brain from external injuries and shocks.

→ An outer covering made of 8 bones is found around the brain, which is called the 'Cranium (Skull)'.

→ Brain related disorders
(i) Alzheimer's → Loss of memory
(ii) Parkinson's → Tremor's in extremities
(iii) Hysteria → Insanity in young women.
(iv) Epilepsy → Having seizures

◉ Parts of Brain.
1. Fore brain
2. Mid brain
3. Hind brain

Brain

- Fore brain
  - Prosencephalon
    - (i) Olfactory lobe
    - (ii) Cerebrum
    - (iii) Diencephalon
      - (a) Thalamus
      - (b) Epithalamus
      - (c) Hypothalamus
- Mid Brain
  - Mesencephalon
    - (i) Corpora Quadrigemina
    - (ii) Cerebral peduncle
- Hind Brain
  - Rhombencephalon
    - (i) Metencephalon
      - Pons
      - Cerebellum
    - (ii) Myelencephalon
      - Medulla oblongata

(1) Fore-brain

(i) Olfactory lobes

→ These are associated with the sense of smell.

Order of development → Scoliodon > Dog > Carnivores > omnivores.

→ It is found in the cerebrum.

(ii) Cerebrum

→ Largest part of the brain.
→ Also called the "intelligent brain"
→ Responsible for thinking and understanding.
→ Controls memory and reasoning.
→ Manages debating and voluntary actions.
→ Shapes personality development.

→ Handles decision-making capacity.

(iii) Diencephalon

(a) Thalamus
↳ Sensation entering the brain and messages existing it cross each other here.
→ It contains control centers for cold, heat, test, touch and external pain.

(b) Hypothalamus
↳ Controls the entire endocrine system.
→ known as the "super commander" of the endocrine system.
→ Regulates body temperature.
→ Contains control centres of emotions.

feelings → Love, anger, malice / Jealousy, Satisfaction, hunger, Thirst, Dancing, Swimming, Sexdrive (Libido).

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(2) Mid-brain
It contains centres for vision and hearing.

(i) Corpora Quadrigemina
↳ 4 solid, round, small structures that are related to the functions of vision and hearing.

(ii) Cerebrual peduncle
↳ fibrous structures that connect the cerebra hemisphere to the other parts.

(3) Hind Brain

(i) Pons varolli → fibrous structure that connects the hind brain to other parts.

→ Here, the respiratory system is controlled by the pneumotoxic center.

(ii) Cerebellum → Second largest part of Brain.
→ Mantains body balance.
→ Hence called the "tree of life".
→ Controls voluntary muscles
→ Controls speech (when, what, how and how much to speak).
Affected by alcohol consumption.

(iii) Medulla oblongata → It contains the control centers for involutary actions.

→ Involuntary actions are of two types ;

Biological action
- These are actions that are necessary for staying alive
- Eg ; Heart beat
  Blood pressure
  Respiration / Breathing
  Excretion
  Digestion

Accessory action
- These are actions that help in staying alive but not essential.
- Eg ; Sneezing
  Hiccuping
  Burping
  Yawning
  Snoring
  Vomitting and

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(ii) Spinal Cord :

→ length → 42-45 cm

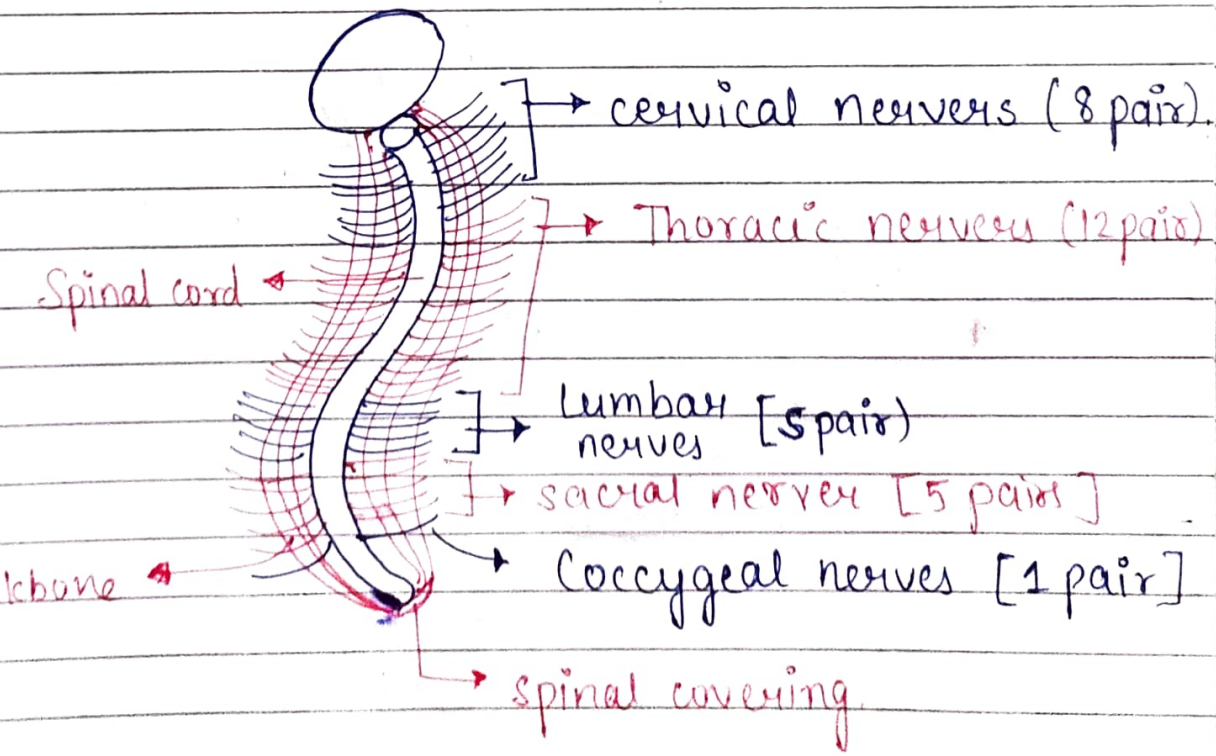
→ It connects to the brain through a hole called the "foramen magnum".

→ Just like the brain, it is also surrounded by three membranes called the meninges.

→ CSF continuously circulates around it, providing nutrition to the spinal cord and helping to remove waste products.

→ It is surrounded by the "vertebral column"/ backbon which is called the "protective shield" of the spinal cord.

→ 31 pairs of spinal nervers are connected to the spinal cord.

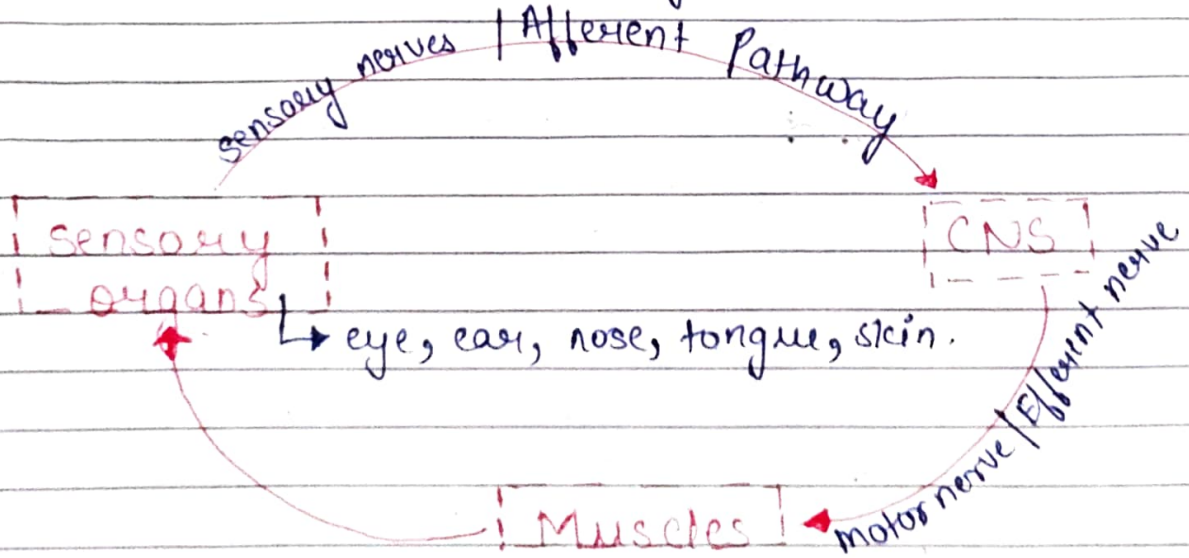


→ Spinal cord controls reflex actions.

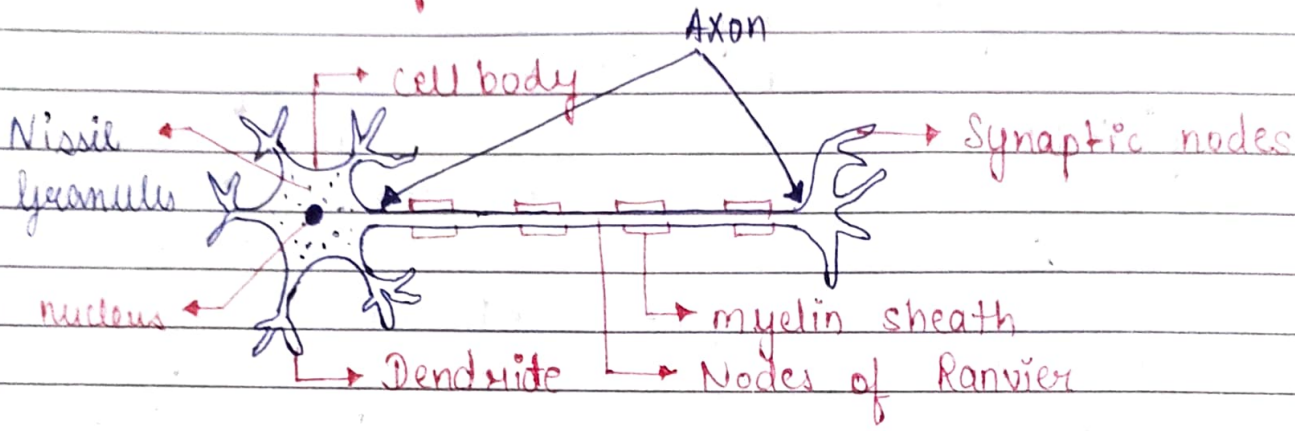
* Reflex actions → Actions that happens automatically without thinking.

→ Marshall hall first explained them.

For eg → (i) Suddenly pulling the hand back upon touching a hot object.
(ii) Salivating in the mouth upon seeing food
(iii) Construction of eyes pupil when going into bright light.
(iv) Sudden lifting of foot upward when a thorn pricks the foot.
(v) Turning the neck towards the direcn of the sound upon hearing a loud noise.



Structure of nerve cell



[Squence of information flow in nerve cell (Neuron)
↳ Dendride → cell body (soma) → Axon → Synaptic nodes

connection b/w 2 neurons → Synaps
→ Synaps is not physical connection, it is chemical connection.

- Speed of nerve impulse transmission → 130-200 m/s
- Ions responsible for nerve impulse transmission
  ↳ Na^+ | K^+

Neurotransmitters → It increase nerve impulse speed.
for eg; Acetylcholine
Epinephrine
Norepinephrine.

Neuroinhibitors → It slow nerve impulse speed.
for eg; GABA
↳ [Gama Amino Butyric Acid]

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② AUTONOMOUS NERVOUS SYSTEM

↳ first explained by langley

Sympathatic

- Source of stress
- Becomes active during times of grief, stress & pain

Func^n (i) Increase heartbeat
(ii) Increase sweat secretion
(iii) Dilates the pupils of eyes
(iv) Suppresses hunger & thirst
(v) Increases blood pressure
(vi) Decrease saliva secretion

Parasympathatic

- Source of pleasure / relaxation
- Becomes active after stress and pain pass.

Func^n (i) Decrease heartbeat
(ii) Decrease sweat secretion
(iii) Constricts the eye pupils
(iv) Increase hunger & thirst
(v) Decrease blood pressure
(vi) Increase saliva secretion

→ Largest neuron of our body → Saigatic.
→ Largest brain neuron → Vagus
→ Smallest neuron → Trochlear (smallest cranial nerve).